

Theoretical vs Practical Motion of a Small-Scale Projectile Subjected to Lorentz Force

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Objective:

This experiment aimed to examine the acceleration that a rolling current-carrying projectile undergoes when subjected to Lorentz force in a magnetic field, and to then compare its acceleration with a theoretical prediction incorporating rolling inertia.

Setup:

Our design included:

- Six stacks of Neodymium N35 Washers measuring 18.8mm x 2.67mm -9.53mm stacked two high. These stacks were arranged linearly with 22mm between the center points on a galvanized steel plate. Magnets were positioned with their north faces pointing upwards (confirmed with a compass). Magnet proximity was prioritized to encourage uniformity in the average magnetic field across the rails
- Copper sheets served as “rails” for the experiment. Sheets were positioned 4mm above the magnets along the length of the washer arrangement. Sheets were spaced 7.5mm apart from each other. Sheet position was accomplished with tape and cardboard inlays which prioritized sheet proximity to magnets whilst avoiding direct contact. Sheets were not in contact with one another and would serve as charged plates until the projectile completed the circuit.
- The projectile was a solid aluminum bar (cylindrical in dimension) measuring 3.40g in mass, 5.9 mm in diameter, and 44.9mm in length.
- The Copper rails were connected to a PSU set to 20.00V. The Voltage and Amperage was measured with a Fluke meter prior to experimental trials and the plates delivered 8.55A at 20.00V.
- On either end of the rails wood blocks and beakers were placed to apply weight to the rail surface. These weights were intended to stop the projectile at the end of the track, and they also served to increase the regularity of the surface that the projectile would travel over (i.e. “flatten” the track).
- A ruler and distance markings were present to measure the position of the projectile during trials
- During trials the projectile would be placed on the track, motionless, and the PSU would be switched on. The projectile would then move along the track and the process would be recorded.
- Five trials were performed and filmed on an iphone camera at 60fps which corresponds to frame time resolution of approximately 0.0167 seconds. Data was extracted from the time of PSU switch actuation in 0.05s intervals until the last interval prior to the projectile striking the wood blocks.

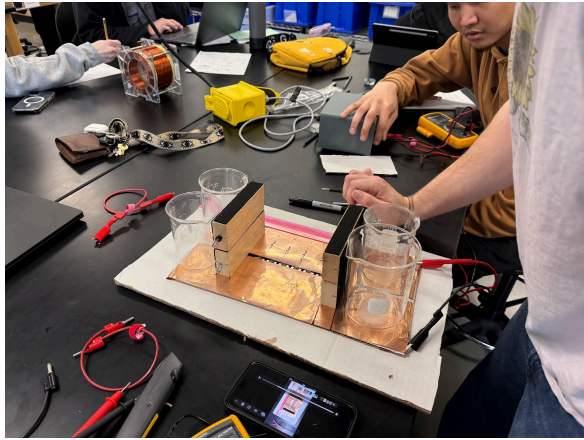


FIG. 1. Pictured experimental setup

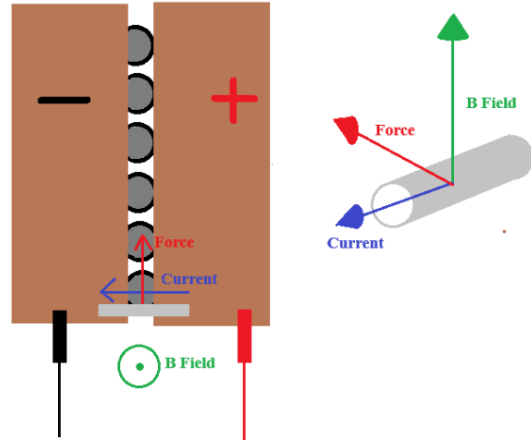


FIG. 2. Diagram indicating Lorentz force direction

After the position-time data was extracted, it was averaged across trials for each time t after the switch was actuated on the PSU. The variability between trials was quantified using the sample standard deviation, and a 95% confidence interval for the mean position was calculated for each time point. These confidence intervals are listed in Fig. 6 and are represented as error bars in the position-time graphs.

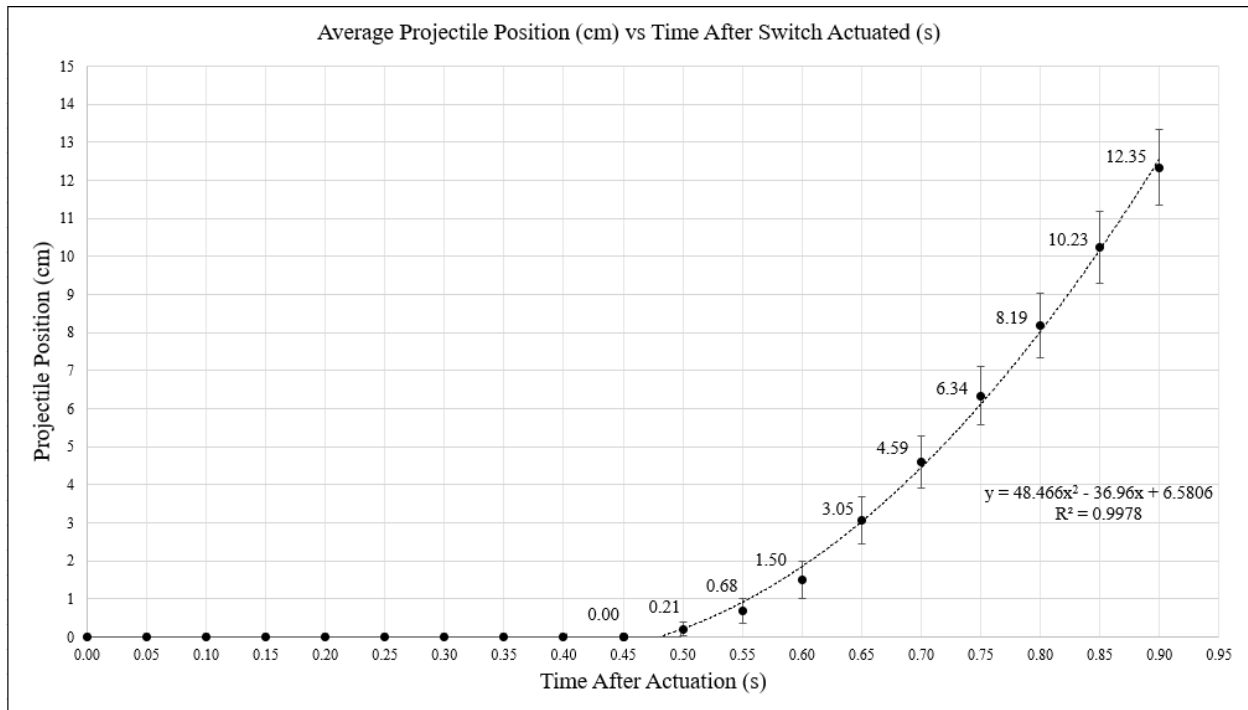


FIG. 3. Graph of projectile position (cm) vs time after PSU switch actuation (s). Error Bars indicate a 95% confidence interval based on the data across the five trials. Labeled numbers are averaged projectile positions (cm) at each given time (s) across the five trials. The trend line was automatically generated in Microsoft Excel to fit the average data under a quadratic model.

To estimate the sensitivity of the calculated projectile acceleration to experimental uncertainty, additional quadratic fits were applied to the upper and lower bounds defined by the 95% confidence interval of the mean position.

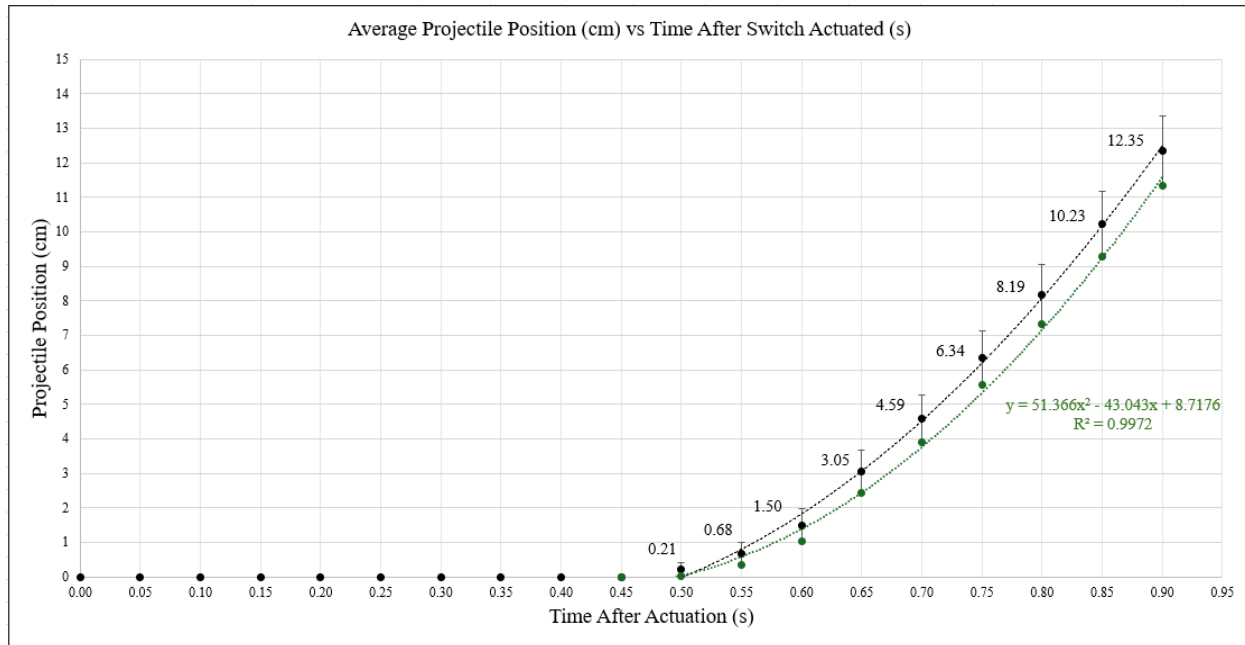


FIG. 4. The same graph as fig. 3 but with the trend line fit to the lower confidence interval

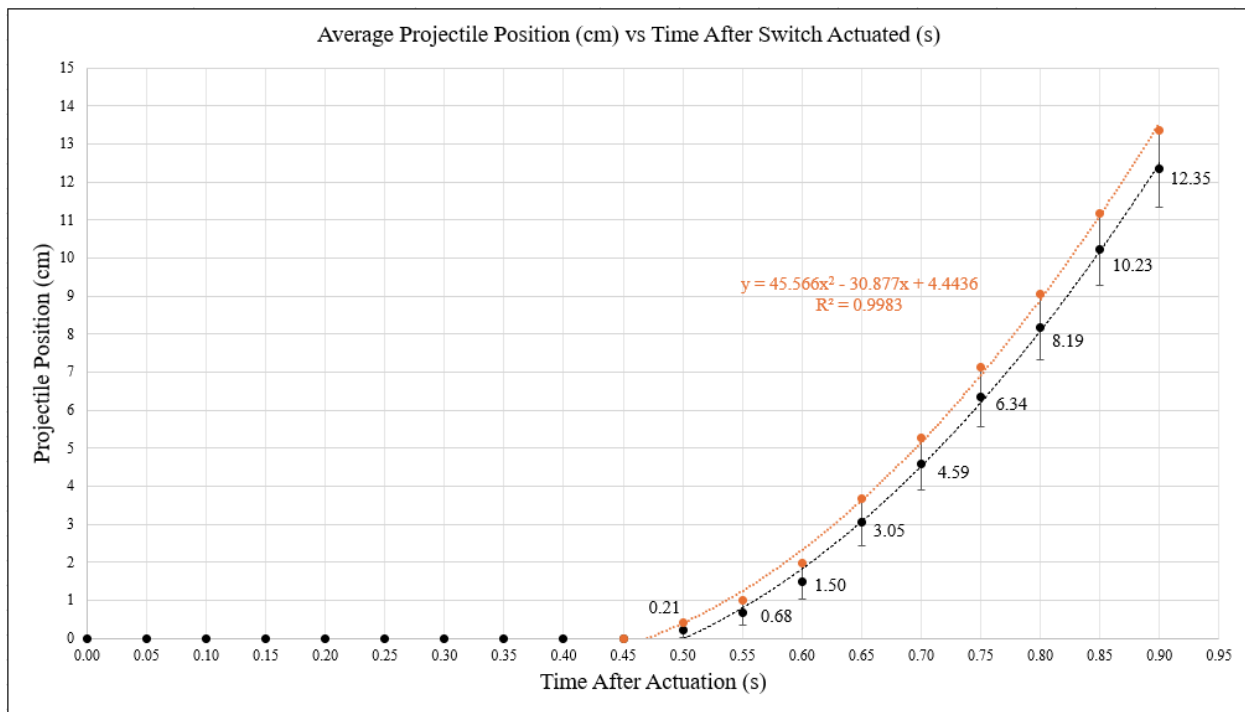


FIG. 5. The same graph as fig. 3 but with the trend line fit to the upper confidence interval

Time After Actuation (s)	<Projectile Position (cm)>	Position STDEV (cm)	Position 95 CI (cm)
0.00	0.00	0.00	N/A
0.05	0.00	0.00	N/A
0.10	0.00	0.00	N/A
0.15	0.00	0.00	N/A
0.20	0.00	0.00	N/A
0.25	0.00	0.00	N/A
0.30	0.00	0.00	N/A
0.35	0.00	0.00	N/A
0.40	0.00	0.00	N/A
0.45	0.00	0.00	N/A
0.50	0.21	0.21	0.19
0.55	0.68	0.37	0.33
0.60	1.50	0.55	0.49
0.65	3.05	0.71	0.62
0.70	4.59	0.78	0.69
0.75	6.34	0.88	0.77
0.80	8.19	0.97	0.85
0.85	10.23	1.09	0.95
0.90	12.35	1.14	1.00

FIG. 6. Corresponding data table for fig. 3-5. It should be noted that <projectile position (cm)> are averages across the five trials for each corresponding time (s). The 95% confidence interval is +/- the listed mean value.

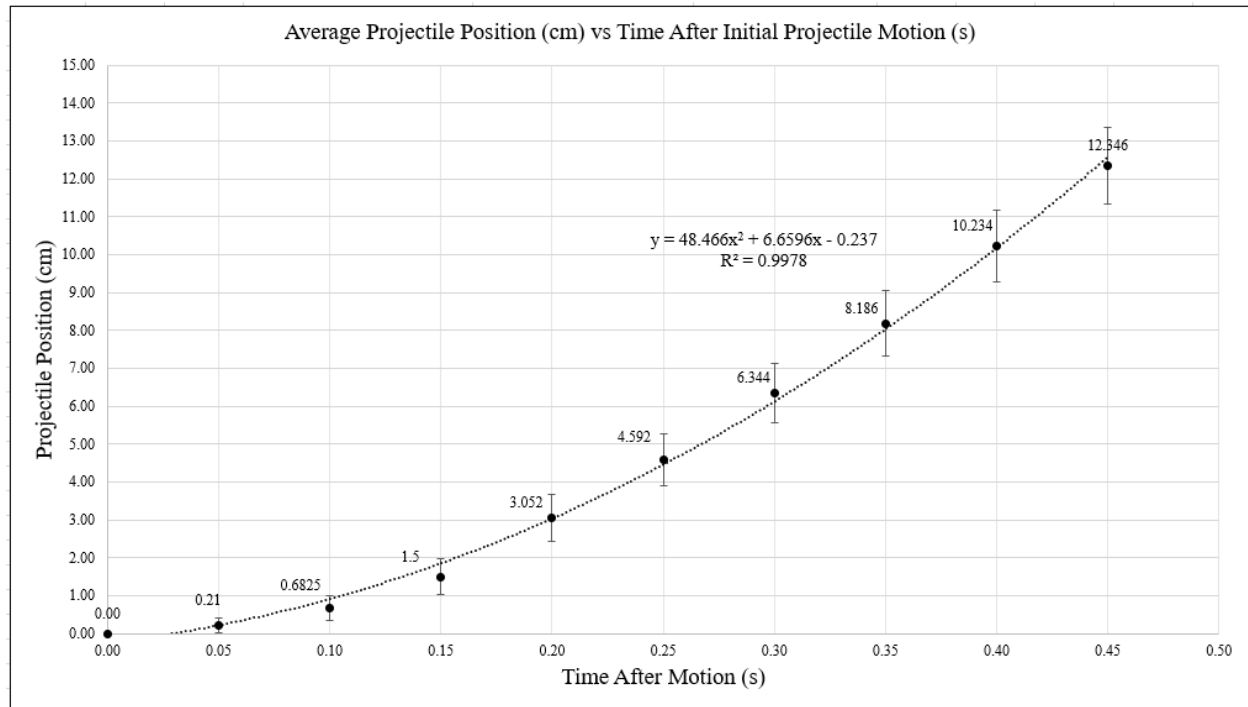


FIG. 7. Graph of projectile position (cm) vs time after initial projectile motion (s). This graph omits startup delay between actuation and initial projectile motion. $t = 0$ is an approximation of the start of projectile motion.

Experimental Calculations:

Variable	Value	Unit
I	8.55	Amp
L	0.0075	Meter
m	0.0034	Kilogram
B (eff)	0.051	Tesla
F (avg)	0.0033	Newton

FIG.8. Table including some measured (row 2-4) and some calculated (row 5-6) variables that may be used in the following calculations.

Using the quadratic kinematic equation for constant acceleration

$$x(t) = x_0 + v_0 t + \frac{1}{2} a t^2$$

And the generated quadratic trend line from fig. 3 (variables exchanged, solving for position initially in cm)

$$x(t) = 48.466x^2 - 36.96x + 6.5806$$

Approximate average experimental acceleration is solvable and yields

$$a \sim 0.97 \frac{m}{s^2}$$

This calculation process is then repeated using the quadratic fits assigned to the upper and lower confidence intervals of the position data. This yields experimental acceleration estimates of 0.903 m/s² and 1.036 m/s², respectively. Thus, the nominal experimental acceleration of 0.969 m/s² is reasonably represented by an uncertainty range of approximately 0.90–1.04 m/s², or 0.97 m/s² ± 0.07 m/s². Using the average acceleration and measured mass in conjunction with Newton's second law provides an average Force

$$F = ma$$

Which yields

$$F_{avg} \sim 0.00330 \text{ N}$$

Now, the Lorentz Force equation:

$$F = ILB$$

Where F is force (N), I is current (A), L is effective conductor length within the magnetic field (m), and B is the magnetic field. Note, for our model, the magnetic field direction is assumed to be 90 degrees from the surface of the magnets. Modifying some variables lets us solve for the average effective B field acting normal to the current and force along the magnet track

$$\frac{F_{avg}}{IL} = B_{eff}$$

Which yields

$$B_{eff} \sim 0.051 \text{ T}$$

To check for physical reasonability, a magnetic field calculator was used.
<https://www.kjmagnetics.com/magnetic-field-calculator.asp?calctype=ring>

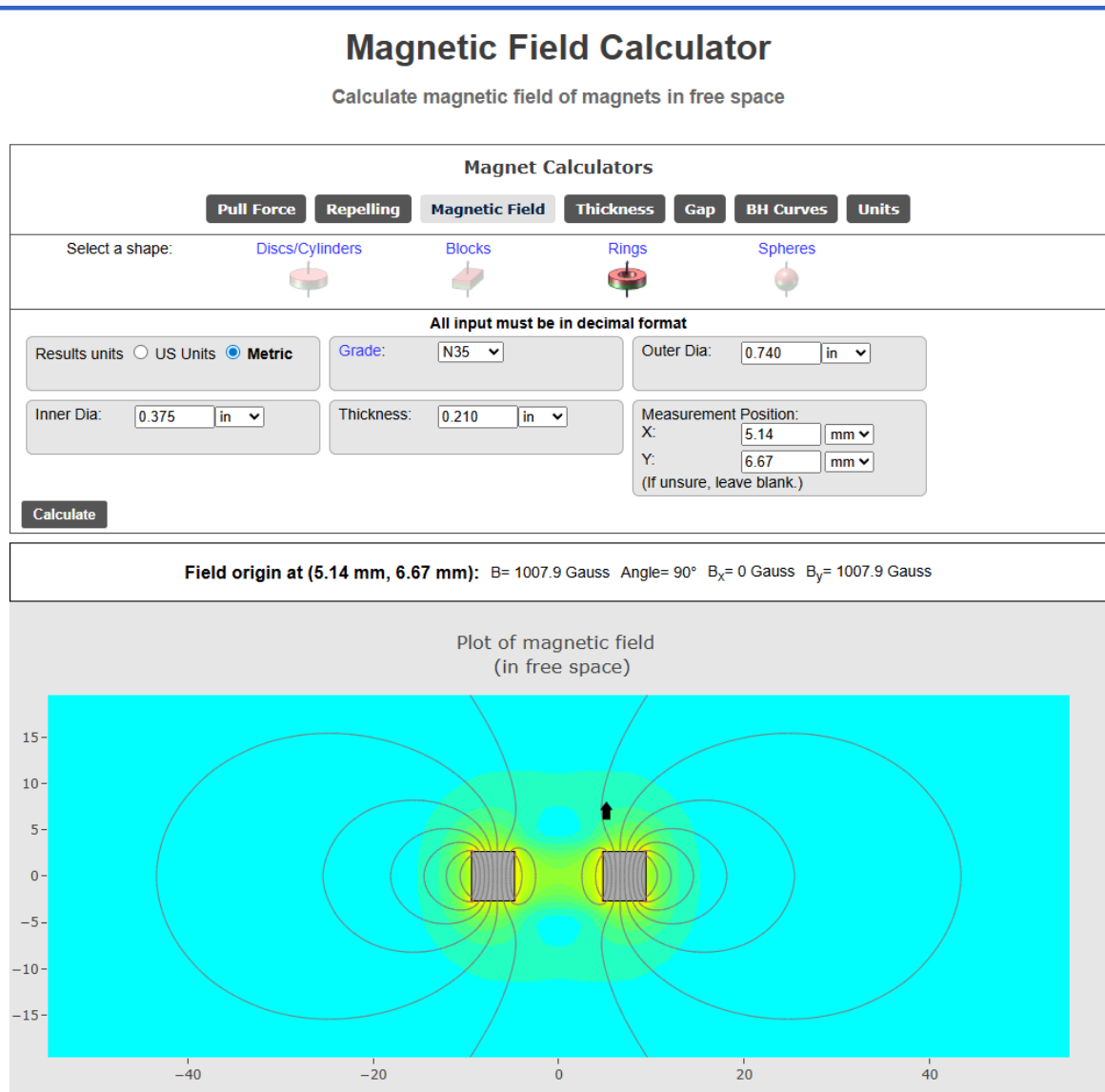


FIG. 9. Magnetic field calculator findings using magnet specifications

The measurement point was selected in order to achieve a perfect 90 degree angle from the surface of the magnet to the direction of the B field. It is also positioned at the same height that our rails sat above our washer magnets. This provides a workable peak B value of 0.10079 T. Due to the sharp falloff of B field strength in the center of the magnet, the irregularity of the direction of the B field towards the outside edges of the washers, and the weakened field in between each magnet, the expected effective magnetic field strength should be significantly lower than the peak of 0.10079T. Therefore, 0.051T is a physically reasonable average B field approximation across the track length.

Theoretical Calculations:

To produce a theoretical model for the physics involved in our model, we started with

$$F = ILB$$

This assumes some constants: L , B , and I . The general equation for a rigid rolling object subject to a force is

$$a = \frac{F}{m + \frac{I_{rot}}{R^2}}$$

Where I (rotation) is the moment of inertia about the center mass and R distance from the object's center of mass to the point of contact with the surface. Substituting for force

$$a = \frac{ILB}{m + \frac{I_{rot}}{R^2}}$$

For a solid cylinder:

$$I_{rot} = \frac{1}{2}mR^2$$

After substituting for I (rotation) into the formula for acceleration, we are left with

$$a = \frac{2ILB}{3m}$$

At this point, our calculated B (eff) will be used in place of the uniform B variable. This means the ideal rolling model will likely adopt some of the non-ideal parameters present in the inference of B (eff). This may lead to an underpredicted acceleration when compared with the measured results.

$$a = \frac{2ILB_{eff}}{3m}$$

When solved, acceleration is predicted at

$$a \sim 0.65 \frac{m}{s^2}$$

Summary and Conclusions:

Overall it appeared our theoretical model did not agree with the experimental result. Average experimental projectile acceleration was determined to be $0.97 \pm 0.07 \text{ m/s}^2$. Using mean values for comparison, the 0.65 m/s^2 theoretical average acceleration is approximately 33% lower than the experimentally measured average acceleration 0.97 m/s^2 . There are several possible reasons. As mentioned previously, the utilization of B_{eff} in the ideal rolling model may introduce the inefficiencies from experiment into the theory prediction, potentially compounding them and leading to an underprediction. Another factor is a potentially overcalculated experimental acceleration from the misassignment of $t = 0$. The exact timing of when the bar began to accelerate is difficult if not impossible to determine given our experimental setup. $t = 0$ is calculated from the measurement of the bar's initial motion extracted from 60fps video, but the bar was likely accelerating prior to movement, just not enough to overcome the system's static friction. Also, current likely built along the rod and was not instantaneously applied, further complicating the acceleration estimate by introducing a non constant acceleration. Small differences in this $t = 0$ timing selection could have lent to an overestimation of acceleration by generously assigning $t = 0$ to the beginning of projectile motion. Despite the discrepancy between theoretical and experimental acceleration, the results demonstrate the expected Lorentz force motion and produce a physically reasonable effective magnetic field estimate of 0.051 T. With improvements to timing precision, rail rigidity, and magnetic field uniformity, the experimental results could likely achieve closer agreement with the theoretical model. More specifically, improvements could include:

- Replacing the cardboard structures with wood or a plastic (generally, a rigid material that is non conductive).
- Using rigid copper instead of malleable copper sheeting for the rails to flatten the surface the projectile rolls across.
- Reducing projectile mass by using a hollow aluminum cylinder.
- Use different magnets/materials to control B field direction effectively. This could include using flat bar magnets with shielding materials to attempt to direct the B field vertically. Bar magnets could also reduce the space between magnets and increase B field uniformity. These factors could allow for a more calculable average B field across the rail length, eliminating the need to estimate an experimental effective B field for the theoretical calculations.
- Use higher framerates for capturing data.
- Attempt to refine $t=0$. This could involve an LED attached to the PSU captured in the video frame that lights up when current is delivered. Also, some sort of electronic switch would be better than the mechanical switch for timing accuracy.